Assignment 2

1.Create a table listing and describing the main types of hackers (such as white hat, black hat, grey hat, script kiddie, hacktivist, and state-sponsored) and give one real-world example or news headline for each type.

Ans-

Type of hacker	Description	Real-world example or headline
White hat	An ethical hacker who tests security with permission and reports vulnerabilities.	Two white hats hack a Tesla, get to keep it — researchers demonstrated a Tesla software vulnerability at Pwn2Own in 2019. WeLiveSecurity
Black hat	A hacker who accesses systems illegally to steal data, cause damage, or make money.	SolarWinds cyberattack — a major cyber-espionage campaign attributed by UK and US authorities to Russia's Foreign Intelligence Service. National Cyber Security Centre
Grey hat	A hacker who may discover a vulnerability without permission, but does not necessarily intend harm. Their actions can still be illegal.	Example: someone finds a security flaw on a public website without authorization and reports it to the owner.
Script kiddie	An inexperienced attacker who uses existing hacking tools or scripts without fully understanding them.	Example: a beginner uses a publicly available tool to attempt a website disruption.
Hacktivist	A hacker or group that uses cyber activity to promote a political or social cause.	Global Revival of Hacktivism Requires Increased Vigilance from Defenders — Google Cloud discusses hacktivist activity and politically motivated cyber operations. Google Cloud Blog
State-sponsored	A hacker or group supported, directed, or funded by a government to pursue objectives such as espionage or disruption.	UK and US call out Russia for SolarWinds compromise — official attribution to Russia's SVR. National Cyber Security Centre

2.Pick any popular app you use (Instagram, Paytm, Zomato, etc.) and describe a scenario for each type of hacker: how would a white hat, black hat, and grey hat hacker interact with this app? Write 2-3 sentences per scenario.

Ans-

A. White hat hacker

A white hat hacker obtains permission to test Instagram's security or works within an authorized bug bounty program. They test for vulnerabilities within the permitted scope and report any findings responsibly so Meta can fix them.

B. Black hat hacker

A black hat hacker might use a phishing page to trick Instagram users into revealing their passwords. They could then misuse compromised accounts, steal private information, or send scam messages. These activities are unauthorized and harmful.

C. Grey hat hacker

A grey hat hacker might discover a security weakness in Instagram without first obtaining permission. They report the issue to Meta, but testing without authorization can still violate laws or service rules, even if their intention is to help.

**3.Given the following user actions, classify each as white hat, black hat, grey hat, or other: reporting a bug to the company, stealing user data, testing your own account for vulnerabilities, defacing a website for a political cause, and participating in a government cyber operation.

Hint: Use your table from Task 1 for quick reference.**

Ans-

No.	User action	Classification	Reason
1	Reporting a bug to the company	White hat (if authorized and responsibly reported)	Helps the company improve security.
2	Stealing user data	Black hat	Unauthorized data theft harms users.
3	Testing your own account for vulnerabilities	Other — authorized self-testing	Testing your own account is generally legitimate when it does not involve systems or data you are not authorized to test.
4	Defacing a website for a political cause	Hacktivist	The action is politically motivated, but unauthorized defacement is still malicious or illegal conduct.

5	Participating in a government cyber operation	State-sponsored / government cyber activity	The category depends on the person's role and whether the operation is government-directed. Not every government operation is automatically malicious.

4.Interview-style: Imagine you are explaining to your non-technical friend the difference between a white hat and a black hat hacker using a cricket match analogy. Write your explanation in 4-5 sentences.

Ans-

A white hat hacker is like a cricket umpire or an authorized coach who checks whether the game follows the rules. They help identify problems and report them so the game becomes fair and safe. A black hat hacker is like a player who secretly tampers with the opponent's equipment to gain an unfair advantage. The white hat works with permission to protect the system, while the black hat breaks rules to cause harm or gain something illegally. Both may understand the game very well, but their permission, intentions, and actions are different.